Exact Center-Contour Charging on a Four-Dimensional Cubical Block

The global cost hidden by a frustrated microscopic conditioning

20 July 2026

Abstract

A center-valued boundary conditioning can make one plaquette adjacent to a selected link frustrated while the other five are aligned. Treating that boundary as free hides its global curvature. We reconstruct the complete center field on the open side-two four-dimensional cubical block. The plaquette complex over $\mathbb{F}_2$ has ranks 136 and 80 on the two adjacent coboundaries and is exact at all 216 plaquettes. An exhaustive syndrome certificate proves that the one-frustrated/five-aligned incident pattern requires at least five negative plaquettes globally; a single negative link attains five. We then derive a volume-uniform rooted contour bound for the center subsector. This is an exact dependent-sector theorem and a calibration of boundary charging, not control of continuous nonabelian fluctuations.

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1 Center cochains and Bianchi exactness

Let C^1, C^2, C^3 be the link, plaquette, and three-cell cochains of the open side-two four-cube, with coefficients in $\mathbb{F}_2$. A center-valued link field is $x \in C^1$; its negative plaquette field is

$$y = d_1 x \in C^2.$$

The cubical Bianchi identity is $d_2 y = 0$.

The block contains 216 links, 216 plaquettes, and 96 three-cells. Exact row reduction gives

$$\operatorname{rank}_{\mathbb{F}_2} d_1 = 136, \qquad \operatorname{rank}_{\mathbb{F}_2} d_2 = 80, \qquad d_2 d_1 = 0. \tag{1}$$

Proposition 1.1 (Exactness at plaquettes)**.** *Every Bianchi-closed center plaquette field on this block is an exact link coboundary:*

$$\ker d_2 = \operatorname{im} d_1.$$

Proof. The inclusion follows from $d_2 d_1 = 0$. By (1),

$$\dim \ker d_2 = 216 - 80 = 136 = \dim \operatorname{im} d_1.$$

The inclusion is therefore equality. □

2 Minimum charge of the frustrated incident pattern

Fix the degree-six link selected in Stage 10. Among its six incident plaquettes require one designated plaquette p_* to be negative and the other five to be positive. The previously constructed assignment has one negative center link and full negative-plaquette support

$$\{7, 17, 21, 22, 23\},$$

of weight five, with $p_* = 21$.

Theorem 2.1 (Exact minimum five)**.** *Every center plaquette field satisfying the incident pattern and the Bianchi identity has Hamming weight at least five. The bound is attained.*

Proof. Represent each of the 216 columns of d_2 as an exact 96-bit integer syndrome. Since p_* is present, all additional plaquettes must have column XOR equal to the syndrome of p_*. The other five incident plaquettes are forbidden. Exact meet-in-the-middle enumeration checks the empty set, every allowed singleton, every allowed pair, and every allowed triple. None has the required XOR. Thus no Bianchi-closed completion has total weight at most four. The displayed five-plaquette coboundary verifies attainment. □

This is a finite exhaustive certificate over integers, rather than a floating-point optimization. Independently, no nonzero exact center cocycle has size one or two: no d_2 column is zero and no two columns have the same syndrome. A corner link has a three-plaquette coboundary, so the global nonempty minimum is exactly three.

3 Center-contour summability

Join two plaquettes when they are faces of a common three-cell. This is the component graph forced by the Bianchi parity constraint: even opposite faces of one three-cell must remain in the same component. The exact maximum degree of this adjacency graph on the block is

$$\Delta_{\mathrm{B}} = 20.$$

Let $q \in (0, 1)$ be the relative weight of a negative center plaquette. In the center-restricted heat-kernel measure, a curvature support X has relative weight $q^{|X|}$. Its connected components are themselves Bianchi cocycles; on the open block they are exact.

The number of connected plaquette sets of size n containing a fixed root is bounded by $\Delta_{\mathrm{B}}^{2(n-1)}$, using a canonical depth-first traversal of a spanning tree. Since nonempty center contours have $n \geq 3$, we obtain:

Theorem 3.1 (Rooted center-contour bound)**.** *If $e^{\alpha} \Delta_{\mathrm{B}}^2 q < 1$, then*

$$\sup_p \sum_{X \ni p} e^{\alpha |X|} |z(X)| \leq \frac{e^{3\alpha} q^3 \Delta_{\mathrm{B}}^4}{1 - e^{\alpha} \Delta_{\mathrm{B}}^2 q}. \tag{2}$$

The particular Stage 10 incident pattern pays at least q^5.

Proof. Discarding the exact-cocycle restriction only enlarges the sum. Insert the connected-set count and sum the geometric series from $n = 3$. The exact minimum-five theorem supplies the final statement. □

4 Calibration and interpretation

For $\alpha = 0.1$ and $\Delta_{\mathrm{B}} = 20$, the sufficient threshold is

$$q < \frac{e^{-0.1}}{20^2} = 0.00226209354509.$$

At the declared calibration $q = 10^{-4}$, the branching ratio is 0.0442068 and the right side of (2) is 2.25965×10^{-7}. The least suppressed realization of the selected frustration pays $q^5 = 10^{-20}$.

At small heat time the endpoint heat-kernel ratio has logarithmic rate $\log q \sim -\pi^2/(2t)$. For $t = 3 \times 10^{-6}$ this rate proxy is -1.644934×10^6; prefactors are irrelevant to that asymptotic statement.

5 Frontier audit

The theorem demonstrates exactly what arbitrary conditioning erased: a local frustrated staple belongs to a larger center-curvature contour. The proof does not imply that every continuous SU(2) collar failure pays five near-antipodal plaquettes. Continuous fields can spread curvature, and noncentral orbit/topology labels require separate estimates. The strongest claim is C2 for the center subsector.